

Supplementary Material for “A Two-Phase Particle Swarm and Variable Neighborhood Search Algorithm for Continuous Wind Farm Layout Optimization”

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Tables, figures, sections, algorithms and equations of this document are numbered S1, S2, . . . ; references without the prefix S (e.g., Table I or (5)) point to the main text. Unless stated otherwise, all results are based on 30 seed-paired runs of 6,030 objective calls, and the method names are those of the main text (PSO: constriction coefficients (8)).

S-I Model Details

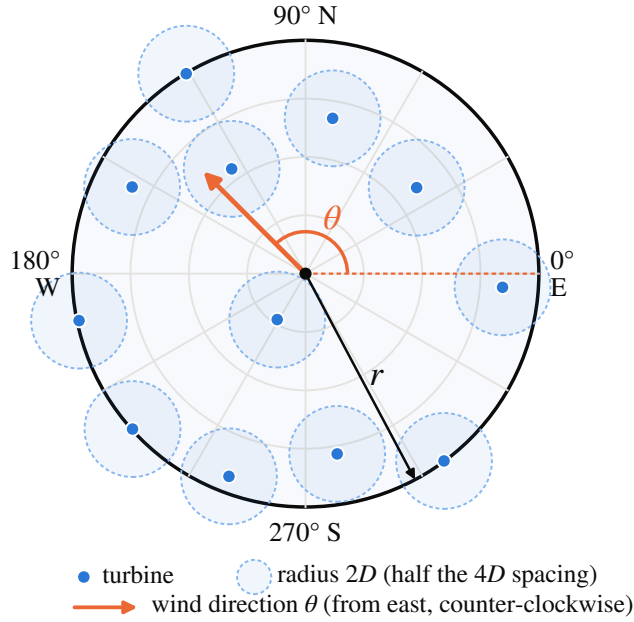


Figure S1: Circular wind farm of radius r . The wind direction θ (direction the wind blows towards) is measured counter-clockwise from east. The points show an optimized layout for 12 turbines in the 750-m farm (Wind Data Set I); the shaded discs have radius $2D$, so two discs touch when their turbines are exactly $4D$ apart.

S-I.A Discrete Form of the Expected Power

Following [1], the integrals of (4) are evaluated with a Riemann sum. The wind direction is divided into bins $\theta_0 = 0^\circ, \theta_1, \dots, \theta_{N_\theta} = 360^\circ$ with probabilities p_l , and the speed range between $s_{\text{cut-in}}$ and s_{rated} into bins $s_0 = s_{\text{cut-in}}, s_1, \dots, s_{N_s} = s_{\text{rated}}$. With $\bar{\theta}_l = (\theta_{l-1} + \theta_l)/2$ and $W_i(s, \theta) = \exp\{-[s/\psi_i(\theta)]^{\zeta(\theta)}\}$,

$$E(P_i) = \sum_{l=1}^{N_\theta} (\theta_l - \theta_{l-1}) p_l \left\{ P_{\text{rated}} W_i(s_{\text{rated}}, \bar{\theta}_l) + \lambda_1 \sum_{q=1}^{N_s} \frac{s_{q-1} + s_q}{2} [W_i(s_{q-1}, \bar{\theta}_l) - W_i(s_q, \bar{\theta}_l)] \right. \\ \left. + \lambda_0 [W_i(s_{\text{cut-in}}, \bar{\theta}_l) - W_i(s_{\text{rated}}, \bar{\theta}_l)] \right\}. \quad (\text{S1})$$

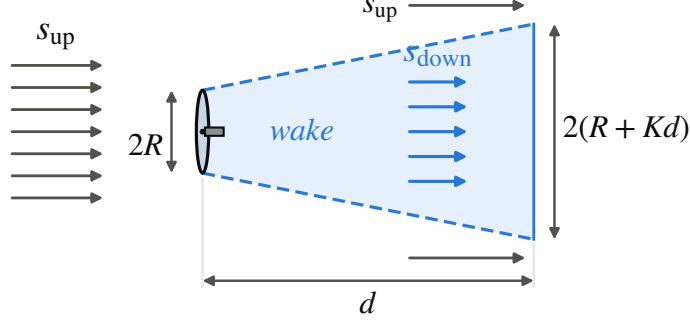


Figure S2: Jensen wake behind a rotor of radius R (schematic, not to scale). The wake widens linearly with the spreading constant K , so that its width at the downstream distance d is $2(R + Kd)$, and the wind speed inside it drops from s_{up} to s_{down} .

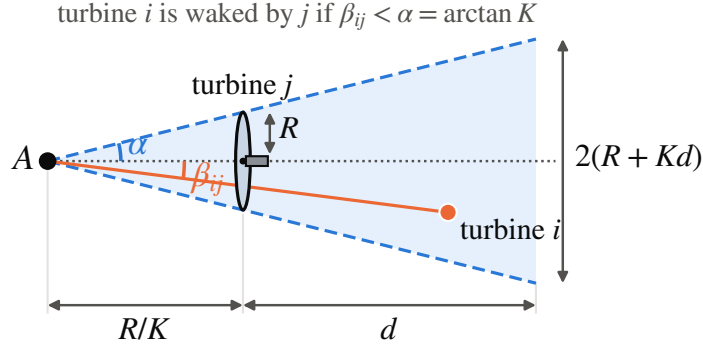


Figure S3: Wake half cone of turbine j (schematic, K exaggerated). The virtual vertex A lies a distance R/K upstream of the rotor, the half-cone angle is $\alpha = \arctan K$, and a downstream turbine i lies in the wake of j when $\beta_{ij} < \alpha$.

Like the earlier studies, we keep the bin-width factor $(\theta_l - \theta_{l-1})$. Because all 24 direction bins are 15° wide and $\sum_l p_l = 1$, the objective equals 15 times the expected farm power in kW, and $\text{AEP} [\text{MWh}] = (\text{objective}/15) \times 8.76$. The linear part of the power curve is -7.0 kW at 3.5 m/s and zero at 3.55 m/s.

S-I.B Gaussian Wake Model Used for the Re-Evaluation

For the re-evaluation of Section VIII-A, the deficit of the Jensen model is replaced by the Gaussian wake model of Bastankhah and Porté-Agel [7],

$$\Delta_{ij} = \left(1 - \sqrt{1 - C_T/(8\sigma^2/D^2)}\right) \exp(-v_{ij}^2/(2\sigma^2)), \quad \sigma/D = K_G d_{ij}/D + \epsilon, \quad (S2)$$

$$\epsilon = 0.2 \sqrt{\frac{1}{2}(1 + \sqrt{1 - C_T})/\sqrt{1 - C_T}},$$

for turbines i downstream of j , with $K_G = 0.04$, the lateral offset v_{ij} and downstream distance d_{ij} of Section II, and the same root-sum-square superposition and Weibull scaling as the benchmark.

S-II Salp Swarm Algorithm and Its Laplacian Variant

Algorithm S1 summarizes SSA [5] and LX-SSA [6] as used in Phase 1 of SSA-VNS and LX-SSA-VNS and as stand-alone methods (Section III-B). A follower moves to the midpoint $\frac{1}{2}(X_m^i + X_m^{i-1})$ between itself and the preceding salp; in LX-SSA it also forms the Laplace candidate (11) and keeps the better of the two, so that one LX-SSA iteration costs $2N_p$ objective calls.

Algorithm S1 SSA and LX-SSA (lines marked LX: LX-SSA only)

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1: Draw and evaluate  $N_p$  salps; sort them and set the best as food source H.
2: for  $t = 1$  to  $T$  do
3:   Update  $\xi_1 = 2 \exp[-(4t/T)^2]$ .
4:   for salp  $i = 1, \dots, N_p$  do
5:     if  $i \leq N_p/2$  then
6:       Update leader  $i$  using (10).
7:     else
8:       Form the follower candidate  $\frac{1}{2}(\mathbf{X}^i + \mathbf{X}^{i-1})$ .
9:       LX: form  $\mathbf{Y}^i$  using (11), evaluate both candidates and keep the better one.
10:    end if
11:  end for
12:  Apply box repair, evaluate  $F_p$  for the population, sort it, and update H.
13: end for
14: return H.

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Table S1: Data Set I, 500-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean among the methods with at least 15 feasible runs. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

N	Ideal	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
2	28091.5	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28086.7 (4.5)
3	42137.2	42131.3 (8.5)	42128.7 (8.8)	42128.5 (10.7)	42114.9 (10.3)	42114.8 (14.2)	42130.7 (8.8)	42131.6 (9.7)	42119.9 (11.0)
4	56182.9	56114.4 (20.7)	56110.0 (25.6)	56101.8 (25.3)	56086.8 (19.8)	56078.8 (22.6)	56092.8 (29.8)	56106.1 (21.7)	56057.0 (55.8)
5	70228.7	70051.2 (28.5)	70053.1 (29.3)	69967.2 (159.2)	69994.0 (34.0)	69972.6 (72.0)	69960.5 (78.6)	69876.9 (199.8)	69876.4 (139.3)
6	84274.4	83918.5 (66.9)	83925.6 (55.8)	83780.8 (213.2)	83535.6 (372.5)	83544.9 (465.3)	83326.7 (761.5)	83299.3 (524.5)	83636.4 (227.9)
7	98320.2	97588.7 (292.3)	97569.6 (325.2)	96529.9 (794.9)	96757.0 (692.4)	96380.6 (1013.8)	95903.3 (1316.3) ¹⁵	96967.1 (590.6)	97049.0 (477.4)
8	112365.9	110777.2 (718.5)	110957.4 (615.4)	109316.4 (1117.4)	109172.6 (1362.4)	108258.3 (1704.3) ²⁸	107632.7 (1592.8) ²	109493.5 (2035.2)	110233.3 (571.9)
9	126411.6	123153.2 (1132.3)	123499.5 (1006.4)	121985.3 (1201.8)	120043.1 (2755.8) ²⁵	119134.2 (2666.4) ²¹	– ⁰	122259.2 (1232.0)	123307.1 (814.5)
10	140457.4	135160.5 (1708.4)	135396.0 (1790.4)	134321.3 (1482.4) ²⁶	132616.3 (2328.0) ¹²	130967.4 (3191.0) ¹³	130994.8 (0.0) ¹	133567.5 (1820.9)	135642.4 (920.1)

Table S2: Data Set I, 750-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean among the methods with at least 15 feasible runs. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

N	Ideal	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
2	28091.5	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.0 (1.6)
3	42137.2	42137.2 (0.0)	42137.0 (1.0)	42137.0 (1.0)	42134.4 (4.1)	42134.2 (4.8)	42136.8 (1.5)	42137.2 (0.0)	42133.3 (5.0)
4	56182.9	56170.9 (10.1)	56170.9 (11.3)	56162.3 (13.6)	56153.8 (17.9)	56149.9 (14.0)	56146.9 (17.5)	56167.0 (13.8)	56139.3 (11.7)
5	70228.7	70169.6 (23.7)	70164.8 (25.1)	70150.9 (25.4)	70122.2 (26.3)	70118.1 (28.6)	70083.9 (31.7)	70125.8 (47.0)	70091.5 (35.7)
6	84274.4	84139.4 (24.3)	84130.1 (30.7)	84095.6 (36.9)	84035.2 (58.4)	84033.5 (58.3)	83948.2 (74.1)	84029.2 (141.3)	83965.7 (117.2)
7	98320.2	98064.7 (54.7)	98059.9 (46.4)	97910.8 (164.8)	97875.2 (169.3)	97840.0 (167.7)	97479.1 (285.1)	97774.4 (232.6)	97663.0 (196.3)
8	112365.9	111921.0 (119.4)	111924.1 (87.8)	111683.9 (282.2)	111506.5 (334.5)	111374.0 (460.8)	110595.8 (445.6)	111411.4 (493.5)	111287.8 (290.8)
9	126411.6	125552.7 (330.8)	125639.1 (260.7)	125207.2 (539.4)	124756.2 (610.5)	124698.0 (960.4)	121674.9 (2276.5) ²³	124780.0 (716.8)	124787.6 (424.2)
10	140457.4	139122.6 (345.7)	139212.9 (276.9)	138304.3 (685.9)	137817.6 (1160.1)	137748.6 (1116.3)	134937.7 (998.5) ⁴	138170.4 (764.7)	138182.1 (569.0)
11	154503.1	152343.5 (565.1)	152626.6 (443.6)	151286.3 (1117.4)	150404.1 (1042.2) ²⁹	149863.8 (1886.6) ²⁹	148555.7 (1134.8) ³	151004.2 (1339.0)	151173.5 (598.0)
12	168548.8	165526.4 (774.4)	165826.3 (766.5)	164279.2 (1284.1)	162903.5 (1685.8)	161630.5 (1988.2) ²⁸	– ⁰	163712.4 (1575.6)	164478.5 (634.3)

Table S3: Data Set I, 1000-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean among the methods with at least 15 feasible runs. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

N	Ideal	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
2	28091.5	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.3 (0.7)
3	42137.2	42137.2 (0.0)	42137.2 (0.0)	42137.2 (0.0)	42136.7 (1.6)	42136.7 (1.4)	42137.2 (0.0)	42137.2 (0.0)	42136.6 (1.8)
4	56182.9	56181.4 (3.1)	56181.8 (3.1)	56180.0 (3.8)	56169.4 (9.2)	56172.9 (7.8)	56166.9 (8.4)	56177.3 (8.6)	56165.3 (10.2)
5	70228.7	70209.8 (10.8)	70208.0 (13.1)	70202.1 (15.9)	70174.0 (19.7)	70183.3 (21.0)	70148.7 (22.4)	70193.3 (24.5)	70159.6 (39.9)
6	84274.4	84214.5 (19.7)	84213.1 (24.2)	84175.9 (36.0)	84149.9 (29.7)	84142.7 (23.5)	84096.1 (24.0)	84170.0 (49.5)	84064.6 (113.6)
7	98320.2	98192.1 (32.2)	98178.1 (38.0)	98128.4 (59.9)	98084.7 (47.0)	98078.4 (82.1)	97942.3 (67.9)	98064.6 (118.7)	97930.4 (157.1)
8	112365.9	112142.9 (48.9)	112125.1 (48.4)	112008.3 (131.2)	111947.1 (125.7)	111924.6 (164.7)	111608.3 (168.9)	111862.5 (246.1)	111736.9 (163.4)
9	126411.6	126003.5 (103.1)	125993.3 (89.8)	125842.0 (277.4)	125761.1 (168.4)	125701.5 (265.4)	124996.3 (282.9)	125589.7 (312.6)	125450.0 (305.1)
10	140457.4	139874.2 (102.1)	139834.3 (108.2)	139561.7 (359.4)	139348.0 (433.0)	139354.2 (388.5)	137969.0 (623.0)	139290.3 (447.7)	138991.7 (381.3)
11	154503.1	153621.0 (192.7)	153621.2 (168.9)	153184.6 (292.3)	152825.1 (385.8)	152736.8 (558.8)	150189.7 (1068.9) ²⁹	153010.5 (393.8)	152538.4 (405.6)
12	168548.8	167262.1 (383.9)	167223.9 (330.8)	166530.1 (760.0)	166091.9 (816.6)	165814.4 (827.9)	162390.4 (1765.9) ¹⁷	166510.6 (669.8)	166171.0 (462.2)
13	182594.6	180623.7 (496.4)	180638.4 (495.0)	179973.5 (642.1)	179101.5 (1028.9)	178913.6 (967.4)	174763.4 (1389.6) ⁴	179762.6 (1091.2)	179275.1 (644.9)
14	196640.3	194092.5 (813.0)	194082.6 (688.3)	193260.8 (939.8)	192235.7 (1504.8)	191046.6 (1754.6)	– ⁰	192874.6 (1745.6)	192657.7 (605.3)
15	210686.1	207308.1 (894.1)	207263.9 (720.0)	207119.3 (639.3)	204777.6 (1420.5)	203681.8 (1899.5) ²⁹	– ⁰	206486.7 (1093.6)	205876.8 (798.9)

S-III Per-Case Results and Additional Tables

Tables S1–S6 give the per-case results of the main comparison; the remaining tables give the minimum-spacing re-optimization, the full case-level statistics, the coordinates of the best layouts, the computational cost and the per-case values of the budget-split, initialization, budget, Horns Rev 1, IEA37 and robustness studies of the main text. All tables in this section are generated by `analysis/mpce_results.py`.

Table S4: Data Set II, 500-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean among the methods with at least 15 feasible runs. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

N	Ideal	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
2	14630.8	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14626.0 (5.6)
3	21946.1	21909.3 (15.9)	21904.5 (13.8)	21907.1 (18.9)	21900.9 (11.1)	21899.7 (9.5)	21901.2 (5.5)	21932.3 (17.2)	21892.0 (45.9)
4	29261.5	29124.8 (63.8)	29142.9 (54.2)	29067.5 (60.7)	29028.7 (93.8)	29004.2 (87.4)	29044.2 (61.5)	29070.8 (94.1)	28993.5 (96.1)
5	36576.9	36100.0 (100.0)	36118.0 (117.9)	35958.0 (130.6)	35918.9 (74.0)	35880.1 (168.6)	35789.6 (193.8)	35878.2 (160.9)	35914.9 (144.5)
6	43892.3	42851.5 (166.9)	42857.9 (163.5)	42638.7 (198.9)	42444.3 (224.5)	42512.1 (227.9)	42257.5 (374.9)	42560.3 (202.5)	42601.8 (178.6)
7	51207.6	49332.5 (195.5)	49337.3 (189.8)	49008.6 (239.7)	48887.9 (304.1)	48626.5 (340.5)	48203.9 (767.8) ¹⁵	49032.1 (314.4)	48959.7 (202.6)
8	58523.0	55727.9 (300.0)	55668.1 (312.6)	55274.9 (503.7)	54900.5 (471.3)	54699.5 (620.7) ²⁸	54451.3 (618.7) ²	55182.8 (738.6)	55427.1 (303.9)
9	65838.4	61754.3 (344.3)	61643.9 (481.7)	61380.0 (541.4)	60581.0 (844.8) ²⁵	59963.3 (652.2) ²¹	– ⁰	61166.8 (574.8)	61516.6 (326.2)
10	73153.8	67773.5 (477.6)	67472.2 (637.3)	67106.6 (779.5) ²⁵	65880.7 (761.6) ¹²	65868.7 (586.0) ¹³	64240.2 (0.0) ¹	67056.0 (749.1)	67297.9 (402.5) ²⁹

Table S5: Data Set II, 750-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean among the methods with at least 15 feasible runs. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

N	Ideal	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
2	14630.8	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.2 (1.8)
3	21946.1	21945.5 (2.5)	21945.0 (3.9)	21946.1 (0.0)	21940.0 (8.2)	21937.3 (8.1)	21942.5 (5.8)	21946.1 (0.0)	21938.3 (11.3)
4	29261.5	29225.9 (22.5)	29221.2 (22.7)	29210.5 (34.5)	29182.5 (36.2)	29201.4 (25.7)	29195.3 (26.2)	29227.3 (29.9)	29175.2 (48.2)
5	36576.9	36473.4 (36.9)	36480.6 (37.9)	36417.1 (67.3)	36332.0 (91.4)	36366.8 (74.3)	36277.7 (105.9)	36389.7 (97.1)	36267.4 (108.2)
6	43892.3	43596.3 (87.7)	43611.5 (96.3)	43500.1 (119.6)	43419.5 (153.1)	43377.4 (175.7)	43142.2 (144.8)	43367.4 (201.2)	43264.0 (140.5)
7	51207.6	50584.2 (207.2)	50592.0 (149.5)	50370.1 (169.6)	50264.8 (202.5)	50095.4 (193.6)	49696.6 (156.3)	50180.2 (303.8)	50174.3 (220.0)
8	58523.0	57411.1 (173.3)	57404.8 (185.9)	57033.9 (298.6)	56806.2 (303.6)	56659.2 (332.0)	56033.1 (331.4)	56855.9 (391.5)	56837.8 (340.9)
9	65838.4	64016.2 (259.2)	63991.1 (213.9)	63632.1 (366.8)	63291.6 (402.5)	63252.7 (342.8)	61970.8 (763.0) ²³	63217.0 (515.4)	63472.4 (272.8)
10	73153.8	70558.3 (205.2)	70484.8 (208.8)	70000.5 (393.6)	69520.3 (512.6)	69480.7 (385.7)	68103.2 (423.7) ⁴	69930.2 (355.0)	70128.5 (305.6)
11	80469.2	76912.3 (421.8)	76804.4 (395.8)	76265.0 (512.1)	75742.5 (677.1) ²⁹	75181.6 (681.1) ²⁹	74596.6 (476.3) ³	76099.7 (566.3)	76420.9 (338.9)
12	87784.5	82968.9 (469.9)	82874.2 (531.0)	82500.8 (492.0)	81822.1 (625.7)	81341.8 (792.1) ²⁸	– ⁰	82368.6 (603.1)	82792.2 (462.3)

Table S6: Data Set II, 1000-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean among the methods with at least 15 feasible runs. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

N	Ideal	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
2	14630.8	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.4 (1.2)
3	21946.1	21946.1 (0.0)	21946.1 (0.0)	21946.1 (0.0)	21945.6 (1.8)	21945.3 (2.8)	21946.1 (0.0)	21946.1 (0.0)	21943.9 (3.7)
4	29261.5	29254.5 (7.9)	29251.3 (10.2)	29247.6 (12.2)	29232.5 (18.5)	29228.0 (14.9)	29225.9 (12.1)	29253.1 (11.7)	29220.8 (35.1)
5	36576.9	36514.3 (28.7)	36519.7 (32.4)	36510.6 (35.3)	36485.9 (37.5)	36468.2 (46.1)	36402.8 (39.4)	36498.4 (49.1)	36426.0 (58.3)
6	43892.3	43776.8 (48.3)	43789.7 (42.7)	43736.8 (52.7)	43266.4 (74.5)	43656.6 (73.3)	43488.5 (69.7)	43686.1 (101.2)	43554.1 (85.6)
7	51207.6	50952.8 (83.2)	50969.1 (79.2)	50800.0 (120.6)	50731.8 (118.8)	50727.3 (119.3)	50442.7 (136.0)	50696.0 (203.8)	50596.6 (137.4)
8	58523.0	58050.4 (138.4)	58051.3 (154.0)	57883.4 (180.0)	57739.3 (142.0)	57645.1 (186.6)	57210.0 (191.8)	57588.6 (311.3)	57528.0 (226.3)
9	65838.4	65052.9 (170.2)	65065.3 (177.0)	64730.4 (233.0)	64476.4 (203.0)	64428.7 (273.1)	63790.1 (227.2)	64430.8 (311.5)	64357.7 (268.5)
10	73153.8	71906.7 (236.9)	71902.4 (238.0)	71541.9 (328.8)	71139.4 (336.8)	71099.4 (266.9)	70190.1 (286.9)	71342.3 (339.4)	71142.5 (257.2)
11	80469.2	78633.2 (289.7)	78478.9 (290.3)	78133.1 (465.6)	77697.5 (425.8)	77561.6 (498.0)	75960.9 (577.6) ²⁹	77948.1 (483.3)	77762.2 (436.7)
12	87784.5	85357.5 (330.7)	85172.5 (311.4)	84647.1 (650.8)	84049.6 (660.3)	83981.2 (515.0)	82211.6 (905.2) ¹⁷	84485.7 (606.8)	84385.5 (437.1)
13	95099.9	91904.0 (428.6)	91685.0 (453.5)	91179.4 (556.4)	90267.0 (482.6)	90303.5 (674.2)	88665.1 (496.0) ⁴	90787.8 (806.9)	90948.3 (429.2)
14	102415.3	98223.0 (410.2)	97889.6 (465.1)	97590.4 (615.0)	96936.5 (669.3)	96229.6 (1019.3)	– ⁰	97079.1 (980.3)	97371.5 (411.6)
15	109730.7	104794.8 (531.9)	104375.6 (501.6)	104035.3 (475.4)	102746.3 (827.8)	102327.4 (838.7) ²⁹	– ⁰	103777.6 (794.4)	103901.2 (406.2)

Table S7: Minimum-spacing re-optimization with the original LX-SSA and SSA code (6,030 calls, 30 seed-paired runs per cell): mean benchmark objective, Wilcoxon signed-rank p for LX-SSA vs. SSA, and the mean LX-SSA wake loss as a percentage of the ideal objective at 4D and 6D. A superscript gives the number of feasible runs when fewer than 30.

	DS	r (m)	N	4D (308 m)			5D (385 m)			6D (462 m)			LX-SSA loss (%)
				LX-SSA	SSA	p	LX-SSA	SSA	p	LX-SSA	SSA	p	
1	500	4	56078.8	56086.8	0.237	56082.4	56085.2	0.700	56075.5	56083.9	0.839	0.19 → 0.19	
1	750	8	111374.0	111506.5	0.245	111154.1	111163.4	0.871	110314.4 ²⁸	110847.5	0.021	0.88 → 1.85	
1	1000	8	111924.6	111947.1	0.887	111953.5	111946.4	0.777	111781.5	111964.7	0.005	0.39 → 0.52	
2	500	4	29004.2	29028.7	0.146	28999.5	29019.6	0.229	28981.8	28971.1	0.612	0.88 → 0.96	
2	750	8	56659.2	56806.2	0.061	56621.0	56673.3	0.584	56533.9 ²⁸	56690.6	0.038	3.18 → 3.47	
2	1000	8	57645.1	57739.3	0.084	57580.7	57655.7	0.140	57475.5	57587.2	0.114	1.50 → 1.79	

Table S8: Largest turbine count for which a layout satisfying the circular boundary and the minimum spacing was constructed (multi-start packing search; constructive lower bound), compared with the largest tested N .

r (m)	4D (308 m)	5D (385 m)	6D (462 m)	Largest tested N
500	13	8	7	10
750	26	19	13	12
1000	42	29	21	15

Table S9: Case-level analysis over the 68 benchmark cases. Ranks of the mean feasible benchmark objective within each case (1 = best); a method with fewer than 15 feasible runs out of 30 in a case is ranked below all other methods, by its number of feasible runs (“<15”: number of such cases). Friedman $\chi_F^2 = 319.3$ (7 d.f.), $p = 4.56 \times 10^{-65}$; Iman–Davenport $F_F = 136.5$, $p = 6.71 \times 10^{-109}$. p_z : Holm-adjusted p of the average-rank z test against PSO-VNS (best-ranked method: PSO-VNS). Because mean-rank post hoc tests depend on the pool of compared methods [8], pairwise two-sided Wilcoxon signed-rank tests on the per-case mean wake losses (%) are also given: p_W (Holm-adjusted over the 7 comparisons), matched-pairs rank-biserial correlation r_{rb} (positive = PSO-VNS better) and mean difference $\overline{\Delta L}$ of the wake loss (percentage points, PSO-VNS minus method, over the cases where both methods have at least 15 feasible runs). “Best”: cases in which the method alone ranks first; “Feas.”: percentage of feasible runs.

Method	Avg. rank	Best	Feas. (%)	<15	p_z	p_W	r_{rb}	$\overline{\Delta L}$ (pp)
PSO-VNS	1.82	29	100.0	0	–	–	–	–
PSO	1.97	25	100.0	0	0.713	0.296	+0.16	–0.018
SSA-VNS	3.45	0	99.6	0	2.04×10^{-4}	5.69×10^{-11}	+1.00	–0.330
VNS	4.40	3	100.0	0	2.42×10^{-9}	1.26×10^{-10}	+0.98	–0.465
SSA	5.38	0	97.6	2	8.32×10^{-17}	4.55×10^{-11}	+1.00	–0.611
MS-SLSQP	5.88	1	100.0	0	1.84×10^{-21}	4.18×10^{-11}	+0.96	–0.435
LX-SSA	6.06	0	96.9	2	3.33×10^{-23}	4.55×10^{-11}	+1.00	–0.765
DE	7.04	0	71.4	18	1.04×10^{-34}	5.59×10^{-11}	+1.00	–0.887

Table S10: Coordinates (m, farm center at the origin) of the best layout of all methods and of the best PSO-VNS layout for the largest N of each farm (6,030 calls); objective in benchmark units.

DS	r (m)	N	Method	Objective	Coordinates (x_i, y_i)
I	500	10	MS-SLSQP	138291.2	(469.2, 168.9); (127.1, -90.8); (-183.2, 21.4); (-365.2, -324.6); (-493.9, 32.0); (-320.1, 380.7); (273.4, 411.2); (-38.6, -356.1); (270.8, -412.5); (-6.8, 274.6)
I	500	10	PSO-VNS	137994.8	(485.3, -32.1); (-346.0, 302.7); (322.3, -382.2); (-376.7, -328.3); (317.9, 385.7); (142.8, 131.7); (-59.3, -101.1); (-46.5, 497.1); (-497.9, -0.5); (-88.7, -476.6)
I	750	12	PSO-VNS	167040.0	(-71.5, 110.7); (192.8, -653.4); (741.7, 104.8); (395.5, 126.3); (65.5, -302.0); (-728.9, 172.1); (185.8, 655.0); (-226.9, 453.5); (-531.3, -385.6); (534.1, 524.5); (542.8, -343.4); (-223.2, -715.4)
I	1000	15	PSO	208908.8	(-466.8, -48.4); (923.5, 23.2); (684.6, 693.4); (-230.6, 303.0); (524.3, -694.5); (137.7, 67.9); (517.2, 119.0); (137.7, 964.0); (153.5, -960.7); (-232.5, -607.3); (675.4, -366.5); (-684.6, -361.8); (-980.6, 75.4); (-48.5, -210.1); (-687.0, 528.7)
I	1000	15	PSO-VNS	208720.0	(594.1, 135.8); (-268.3, -387.2); (786.8, -517.5); (-79.4, 263.7); (960.7, 115.7); (-852.8, -274.3); (-998.2, 26.9); (104.5, -420.5); (-552.9, -78.4); (342.1, -881.7); (97.7, 888.8); (-84.5, -912.6); (-515.9, 610.4); (-707.3, 234.6); (355.1, 465.8)
II	500	10	PSO-VNS	68621.3	(499.3, -17.9); (-357.0, 305.5); (302.3, -322.3); (-362.6, -295.2); (336.9, 352.5); (46.0, 150.6); (-45.5, -150.2); (-65.8, 495.6); (-499.5, -2.6); (-102.9, -475.6)
II	750	12	PSO-VNS	83996.8	(-159.3, -214.1); (492.9, 564.5); (-391.5, 639.1); (121.1, -739.2); (-44.2, 619.1); (-662.7, 13.0); (-641.7, -309.1); (-465.3, 325.3); (-348.1, -575.4); (665.0, -320.3); (98.2, 250.2); (724.3, 15.3)
II	1000	15	PSO-VNS	105675.2	(-19.0, 118.7); (-84.9, -328.5); (-602.6, 187.3); (639.7, -523.0); (-942.8, -326.2); (216.5, -974.4); (928.9, -325.3); (-386.7, 906.8); (349.5, 906.0); (-310.8, 528.7); (993.7, 2.3); (322.4, -676.0); (469.3, 84.1); (701.3, 536.4); (-435.2, -798.8)

Table S11: Computational cost at 6,030 objective calls: mean wall-clock time per objective call (including the optimizer overhead) and range over N of the mean time per run (single core). Times come from the machine that produced each result file.

Method	ms per call	s per run
PSO-VNS	0.349	1.5–3.2
PSO	0.244	1.0–2.2
SSA-VNS	0.388	1.7–3.4
SSA	0.399	1.7–3.6
LX-SSA	0.396	1.7–3.6
DE	0.467	2.0–4.3
VNS	0.371	1.6–3.4
MS-SLSQP	1.820	2.6–21.7

Table S12: Sensitivity of PSO-VNS to the budget split between the PSO and VNS phases (25%, 50% and 75% of the 6,030 calls for PSO): mean benchmark objective of the feasible runs (superscript: feasible runs when fewer than 30) and Holm-adjusted Wilcoxon signed-rank p of the 50% split against the 25% and 75% splits (30 seed-paired runs).

DS	r	N	25%	50%	75%	p (25)	p (75)
I	500	6	83851.3	83918.5	83929.5	0.023	0.465
I	500	10	135131.6	135160.5	135421.8	0.556	0.069
I	750	8	111835.7	111921.0	111944.0	0.014	0.045
I	750	12	164894.0	165526.4	165991.2	0.004	0.004
I	1000	10	139700.6	139874.2	139883.7	0.002	0.529
I	1000	15	207367.9	207308.1	207724.0	0.570	3.20×10^{-5}
II	500	6	42809.4	42851.5	42883.5	0.110	0.110
II	500	10	67624.1	67773.5	67799.8	0.115	0.903
II	750	8	57358.0	57411.1	57448.4	0.121	0.121
II	750	12	82784.6	82968.9	83182.0	0.070	0.006
II	1000	10	71769.3	71906.7	71976.0	0.015	0.013
II	1000	15	104349.6	104794.8	104784.9	4.63×10^{-4}	0.968

Table S13: Horns Rev 1 site case, 16-turbine block (Vestas V80 power and thrust curves, measured 12-sector wind climate, installed outline; Jensen wake $k = 0.04$; minimum spacing $4D = 320$ m; wake-free AEP 149.26 GWh/yr): AEP in GWh/yr, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls, mean wake loss (%), feasible runs, Holm-adjusted Wilcoxon signed-rank p of PSO-VNS vs. each method and rank-biserial r_{rb} (positive = PSO-VNS better); run-level Friedman $p = 6.64 \times 10^{-28}$ (over the seeds common to all methods).

Layout / method	AEP	Loss (%)	Feas.	p_{Holm}	r_{rb}
Installed layout	139.51	6.53	–	–	–
PSO-VNS	138.73 (0.83)	7.05	30/30	–	–
SSA-VNS	137.98 (0.71)	7.56	30/30	6.67×10^{-4}	+0.68
LX-SSA-VNS	137.77 (0.91)	7.70	30/30	6.26×10^{-4}	+0.72
RS-VNS	137.96 (0.70)	7.57	28/30	2.24×10^{-4}	+0.78
VNS	137.67 (1.02)	7.77	30/30	2.24×10^{-4}	+0.78
PSO	138.06 (1.15)	7.51	19/30	1.38×10^{-5}	+0.88
SSA	136.66 (1.00)	8.44	30/30	2.79×10^{-7}	+0.97
LX-SSA	136.46 (0.90)	8.58	28/30	1.49×10^{-7}	+0.98
DE	–	–	0/30	1.68×10^{-8}	+1.00
MS-SLSQP	136.80 (0.77)	8.35	30/30	1.49×10^{-7}	+0.98

Table S14: Feasibility-preserving versus uniform random initialization at 6,030 calls on the six largest benchmark cases and the Horns Rev 16-turbine block (7 cases, 30 seed-paired runs each): mean wake loss (%) of the feasible runs averaged over the cases, percentage of feasible runs, and number of cases in which feasible initialization is significantly better / not different / worse (Wilcoxon signed-rank, Holm-adjusted over the methods of each case). Per-case values: Supplementary Table S15.

Method	Loss (%)		Feas. (%)		W/T/L
	random	feasible	random	feasible	
PSO-VNS	4.509	4.370	100.0	100.0	1/6/0
PSO	4.656	5.955	94.8	100.0	0/2/5
SSA-VNS	5.090	4.389	95.7	100.0	6/1/0
SSA	6.183	5.367	82.9	100.0	5/2/0
LX-SSA	6.687	5.464	80.0	100.0	7/0/0
DE	9.461	5.955	1.0	100.0	7/0/0
VNS	5.352	4.248	100.0	100.0	7/0/0
MS-SLSQP	5.069	4.832	99.5	100.0	1/6/0
LX-SSA-VNS	5.216	4.396	99.0	100.0	6/1/0

Table S15: Feasible versus random initialization per case (6,030 calls): mean wake loss (%; Horns Rev: mean AEP in GWh/yr) of the feasible runs, feasible runs, Holm-adjusted Wilcoxon signed-rank p (over the methods of the case) and rank-biserial r_{rb} (positive = feasible initialization better).

Method	Random	Feasible	Feas. (R)	Feas. (F)	p_{Holm}	r_{rb}
<i>Data Set I, $r = 500$ m, $N = 10$</i>						
PSO-VNS	3.771	2.871	30/30	30/30	0.008	+0.59
PSO	3.604	3.698	30/30	30/30	0.477	-0.15
SSA-VNS	4.369	2.932	26/30	30/30	2.76×10^{-6}	+0.91
SSA	5.582	3.611	12/30	30/30	9.31×10^{-8}	+0.98
LX-SSA	6.756	3.649	13/30	30/30	2.98×10^{-8}	+1.00
DE	6.737	3.698	1/30	30/30	1.68×10^{-8}	+1.00
VNS	4.905	2.782	30/30	30/30	2.98×10^{-8}	+1.00
MS-SLSQP	3.428	2.894	30/30	30/30	0.001	+0.70
LX-SSA-VNS	4.888	2.860	29/30	30/30	2.98×10^{-8}	+1.00
<i>Data Set I, $r = 750$ m, $N = 12$</i>						
PSO-VNS	1.793	1.909	30/30	30/30	0.339	-0.20
PSO	1.615	3.252	30/30	30/30	1.68×10^{-8}	-1.00
SSA-VNS	2.533	1.947	30/30	30/30	0.004	+0.66
SSA	3.349	2.623	30/30	30/30	0.006	+0.63
LX-SSA	4.105	2.695	28/30	30/30	8.34×10^{-6}	+0.90
DE	nan	3.252	0/30	30/30	1.68×10^{-8}	+1.00
VNS	2.869	1.778	30/30	30/30	9.59×10^{-5}	+0.83
MS-SLSQP	2.415	2.169	30/30	30/30	0.110	+0.40
LX-SSA-VNS	2.431	1.911	30/30	30/30	0.001	+0.72
<i>Data Set I, $r = 1000$ m, $N = 15$</i>						
PSO-VNS	1.603	1.586	30/30	30/30	0.598	-0.11
PSO	1.624	3.162	30/30	30/30	1.68×10^{-8}	-1.00
SSA-VNS	1.693	1.561	30/30	30/30	0.328	+0.29
SSA	2.804	2.493	30/30	30/30	0.213	+0.42
LX-SSA	3.325	2.619	29/30	30/30	0.004	+0.68
DE	nan	3.162	0/30	30/30	1.68×10^{-8}	+1.00
VNS	1.993	1.541	30/30	30/30	0.001	+0.74
MS-SLSQP	2.283	2.092	30/30	30/30	0.213	+0.42
LX-SSA-VNS	1.777	1.585	30/30	30/30	0.220	+0.38
<i>Data Set II, $r = 500$ m, $N = 10$</i>						
PSO-VNS	7.355	7.411	30/30	30/30	0.761	-0.07
PSO	7.767	9.147	30/30	30/30	1.89×10^{-6}	-0.93
SSA-VNS	8.266	7.401	25/30	30/30	1.49×10^{-4}	+0.79
SSA	9.942	8.831	12/30	30/30	3.26×10^{-7}	+0.97
LX-SSA	9.959	8.924	13/30	30/30	1.04×10^{-7}	+0.98
DE	12.185	9.147	1/30	30/30	1.68×10^{-8}	+1.00
VNS	8.336	7.263	30/30	30/30	2.76×10^{-6}	+0.91
MS-SLSQP	8.005	7.819	29/30	30/30	0.248	+0.32
LX-SSA-VNS	8.548	7.430	29/30	30/30	2.36×10^{-6}	+0.92
<i>Data Set II, $r = 750$ m, $N = 12$</i>						
PSO-VNS	5.486	5.304	30/30	30/30	0.192	+0.35
PSO	5.594	7.395	30/30	30/30	1.68×10^{-8}	-1.00
SSA-VNS	6.019	5.385	30/30	30/30	7.15×10^{-6}	+0.90
SSA	6.792	6.475	30/30	30/30	0.165	+0.40
LX-SSA	7.339	6.624	28/30	30/30	0.003	+0.68
DE	nan	7.395	0/30	30/30	1.68×10^{-8}	+1.00
VNS	6.170	5.059	30/30	30/30	1.79×10^{-6}	+0.94
MS-SLSQP	5.687	5.646	30/30	30/30	0.919	-0.02
LX-SSA-VNS	5.946	5.365	30/30	30/30	1.73×10^{-4}	+0.80
<i>Data Set II, $r = 1000$ m, $N = 15$</i>						
PSO-VNS	4.498	4.549	30/30	30/30	0.556	-0.13
PSO	4.880	6.639	30/30	30/30	1.68×10^{-8}	-1.00
SSA-VNS	5.190	4.504	30/30	30/30	4.99×10^{-6}	+0.91
SSA	6.365	5.648	30/30	30/30	9.38×10^{-4}	+0.72
LX-SSA	6.747	5.775	29/30	30/30	1.53×10^{-6}	+0.94
DE	nan	6.639	0/30	30/30	1.68×10^{-8}	+1.00
VNS	5.425	4.439	30/30	30/30	1.30×10^{-7}	+0.98
MS-SLSQP	5.313	5.145	30/30	30/30	0.192	+0.35
LX-SSA-VNS	5.220	4.559	30/30	30/30	1.99×10^{-4}	+0.79
<i>Horns Rev 1, 16 turbines</i>						
PSO-VNS	138.73	138.88	30/30	30/30	0.560	+0.21
PSO	138.06	136.74	19/30	30/30	0.560	+0.23
SSA-VNS	137.98	138.83	30/30	30/30	2.13×10^{-4}	+0.81
SSA	136.66	137.48	30/30	30/30	0.007	+0.63
LX-SSA	136.46	137.38	28/30	30/30	0.004	+0.68
DE	nan	136.74	0/30	30/30	1.68×10^{-8}	+1.00
VNS	137.67	139.00	30/30	30/30	6.40×10^{-5}	+0.85
MS-SLSQP	136.80	137.23	30/30	30/30	0.135	+0.42
LX-SSA-VNS	137.77	138.72	30/30	30/30	3.36×10^{-4}	+0.78

Table S16: Budget scaling (random initialization) on the six largest benchmark cases and the Horns Rev 16-turbine block: average rank over the 7 cases (ranking rule of Table I), mean wake loss (%) of the feasible runs averaged over the six benchmark cases in which the method has at least half of its runs feasible, and percentage of feasible runs. Per-case values: Table S17.

Method	6,030 calls			30,030 calls			120,030 calls		
	Rank	Loss	Feas.	Rank	Loss	Feas.	Rank	Loss	Feas.
PSO-VNS	1.43	4.084	100	1.14	3.336	100	2.43	3.145	100
PSO	1.86	4.181	95	2.57	3.512	99	6.00	3.419	100
SSA-VNS	4.14	4.678	96	4.43	3.715	100	4.29	3.228	100
SSA	8.29	4.828	83	8.57	4.809	98	8.86	4.193	100
LX-SSA	8.71	5.379	80	9.71	5.407	95	10.00	4.887	99
DE	10.00	–	1	5.86	4.023	81	3.29	3.307	100
VNS	6.43	4.950	100	4.43	3.737	100	2.86	3.174	100
MS-SLSQP	4.43	4.522	100	6.71	3.944	100	7.14	3.584	100
LX-SSA-VNS	5.14	4.802	99	5.57	3.845	100	5.86	3.306	100
RS-VNS	4.57	4.830	93	6.00	3.863	100	4.29	3.255	100

Table S17: Budget scaling per case: mean wake loss (%; Horns Rev: mean AEP in GWh/yr) of the feasible runs and feasible runs at each budget (random initialization).

Method	6,030	Feas.	30,030	Feas.	120,030	Feas.
<i>Data Set I, $r = 500$ m, $N = 10$ (wake loss, %)</i>						
PSO-VNS	3.771	30/30	2.506	30/30	2.343	30/30
PSO	3.604	30/30	2.951	30/30	2.890	30/30
SSA-VNS	4.369	26/30	2.978	30/30	2.305	30/30
SSA	5.582	12/30	4.799	28/30	3.948	30/30
LX-SSA	6.756	13/30	5.618	25/30	4.761	29/30
DE	6.737	1/30	3.111	30/30	3.077	30/30
VNS	4.905	30/30	3.078	30/30	2.142	30/30
MS-SLSQP	3.428	30/30	2.645	30/30	2.088	30/30
LX-SSA-VNS	4.888	29/30	3.416	30/30	2.494	30/30
RS-VNS	4.879	24/30	3.344	30/30	2.278	30/30
<i>Data Set I, $r = 750$ m, $N = 12$ (wake loss, %)</i>						
PSO-VNS	1.793	30/30	1.214	30/30	1.053	30/30
PSO	1.615	30/30	1.241	30/30	1.190	30/30
SSA-VNS	2.533	30/30	1.621	30/30	1.316	30/30
SSA	3.349	30/30	2.316	30/30	1.801	30/30
LX-SSA	4.105	28/30	2.556	30/30	2.203	30/30
DE	–	0/30	1.188	30/30	1.066	30/30
VNS	2.869	30/30	1.663	30/30	1.184	30/30
MS-SLSQP	2.415	30/30	1.872	30/30	1.543	30/30
LX-SSA-VNS	2.431	30/30	1.578	30/30	1.352	30/30
RS-VNS	2.404	30/30	1.711	30/30	1.339	30/30
<i>Data Set I, $r = 1000$ m, $N = 15$ (wake loss, %)</i>						
PSO-VNS	1.603	30/30	1.160	30/30	1.048	30/30
PSO	1.624	30/30	1.195	30/30	1.124	30/30
SSA-VNS	1.693	30/30	1.410	30/30	1.146	30/30
SSA	2.804	30/30	2.160	30/30	1.564	30/30
LX-SSA	3.325	29/30	2.489	30/30	2.018	30/30
DE	–	0/30	2.513	21/30	1.057	30/30
VNS	1.993	30/30	1.390	30/30	1.131	30/30
MS-SLSQP	2.283	30/30	1.844	30/30	1.608	30/30
LX-SSA-VNS	1.777	30/30	1.477	30/30	1.192	30/30
RS-VNS	1.946	30/30	1.496	30/30	1.175	30/30
<i>Data Set II, $r = 500$ m, $N = 10$ (wake loss, %)</i>						
PSO-VNS	7.355	30/30	6.750	30/30	6.434	30/30
PSO	7.767	30/30	6.979	30/30	6.861	30/30
SSA-VNS	8.266	25/30	7.097	30/30	6.488	30/30
SSA	9.942	12/30	8.300	28/30	7.759	30/30
LX-SSA	9.959	13/30	9.202	25/30	8.650	29/30
DE	12.185	1/30	6.874	30/30	6.789	30/30
VNS	8.336	30/30	7.011	30/30	6.461	30/30
MS-SLSQP	8.005	29/30	7.283	30/30	6.986	30/30
LX-SSA-VNS	8.548	29/30	7.101	30/30	6.561	30/30
RS-VNS	8.652	24/30	7.196	30/30	6.645	30/30
<i>Data Set II, $r = 750$ m, $N = 12$ (wake loss, %)</i>						
PSO-VNS	5.486	30/30	4.525	30/30	4.292	30/30
PSO	5.594	30/30	4.628	30/30	4.500	30/30
SSA-VNS	6.019	30/30	4.870	30/30	4.356	30/30
SSA	6.792	30/30	6.005	30/30	5.279	30/30
LX-SSA	7.339	28/30	6.486	30/30	5.952	30/30
DE	–	0/30	4.783	30/30	4.269	30/30
VNS	6.170	30/30	4.953	30/30	4.295	30/30
MS-SLSQP	5.687	30/30	5.173	30/30	4.837	30/30
LX-SSA-VNS	5.946	30/30	4.885	30/30	4.375	30/30
RS-VNS	5.888	30/30	4.919	30/30	4.410	30/30
<i>Data Set II, $r = 1000$ m, $N = 15$ (wake loss, %)</i>						
PSO-VNS	4.498	30/30	3.862	30/30	3.699	30/30
PSO	4.880	30/30	4.079	30/30	3.951	30/30
SSA-VNS	5.190	30/30	4.312	30/30	3.754	30/30
SSA	6.365	30/30	5.276	30/30	4.809	30/30
LX-SSA	6.747	29/30	6.091	30/30	5.740	30/30
DE	–	0/30	5.671	21/30	3.584	30/30
VNS	5.425	30/30	4.326	30/30	3.832	30/30
MS-SLSQP	5.313	30/30	4.848	30/30	4.440	30/30
LX-SSA-VNS	5.220	30/30	4.616	30/30	3.860	30/30
RS-VNS	5.210	30/30	4.510	30/30	3.681	30/30
<i>Horns Rev 1, 16 turbines (AEP, GWh/yr)</i>						
PSO-VNS	138.73	30/30	139.62	30/30	139.63	10/10
PSO	138.06	19/30	139.09	28/30	139.33	10/10
SSA-VNS	137.98	30/30	138.57	30/30	139.59	10/10
SSA	136.66	30/30	137.12	30/30	137.90	10/10
LX-SSA	136.46	28/30	136.60	30/30	136.78	10/10
DE	–	0/30	138.40	8/30	139.80	10/10
VNS	137.67	30/30	139.23	30/30	140.13	10/10
MS-SLSQP	136.80	30/30	137.64	30/30	138.43	10/10
LX-SSA-VNS	137.77	30/30	138.73	30/30	139.54	10/10
RS-VNS	137.96	28/30	138.84	30/30	139.96	10/10

Table S18: IEA Wind Task 37 Case Study 1 (16- and 36-turbine scenarios; official AEP model): AEP (MWh) mean, best and SD over the feasible runs, mean wake loss relative to the wake-free AEP (%), feasible runs, rank (ranking rule of Table I) and Holm-adjusted Wilcoxon signed-rank p of PSO-VNS vs. each method, per scenario and budget.

Method	Mean	Best	SD	Loss (%)	Feas.	Rank	p_{Holm}
<i>IEA37 CS1, 16 turbines, 6,030 calls</i>							
PSO-VNS	395,032	402,782	4,587	15.87	30/30	1	–
PSO	390,540	403,753	7,192	16.82	30/30	2	7.15×10^{-6}
SSA-VNS	388,563	403,251	8,564	17.25	30/30	3	3.80×10^{-4}
VNS	387,637	400,169	5,591	17.44	30/30	4	4.61×10^{-5}
LX-SSA-VNS	386,930	397,429	6,911	17.59	30/30	5	4.61×10^{-5}
RS-VNS	386,206	398,682	5,817	17.75	30/30	6	4.61×10^{-5}
MS-SLSQP	382,605	405,434	20,279	18.51	18/30	7	6.91×10^{-5}
SSA	378,073	395,150	9,076	19.48	30/30	8	6.52×10^{-8}
LX-SSA	373,977	393,285	7,867	20.35	30/30	9	1.68×10^{-8}
DE	338,611	357,391	8,837	27.88	27/30	10	1.68×10^{-8}
<i>IEA37 CS1, 16 turbines, 30,030 calls</i>							
PSO-VNS	402,583	413,332	4,678	14.26	30/30	1	–
PSO	401,347	412,702	4,764	14.52	30/30	2	0.006
LX-SSA-VNS	399,197	407,330	4,330	14.98	30/30	3	0.006
SSA-VNS	396,921	406,925	6,267	15.47	30/30	4	4.42×10^{-4}
VNS	396,192	404,076	4,928	15.62	30/30	5	7.99×10^{-5}
RS-VNS	396,163	408,205	4,977	15.63	30/30	6	8.52×10^{-6}
MS-SLSQP	394,567	410,350	15,316	15.97	29/30	7	0.006
SSA	392,876	401,459	4,394	16.33	30/30	8	4.30×10^{-7}
LX-SSA	390,267	398,770	5,052	16.88	30/30	9	1.68×10^{-8}
DE	359,952	397,403	12,352	23.34	30/30	10	2.98×10^{-8}
<i>IEA37 CS1, 36 turbines, 6,030 calls</i>							
VNS	741,566	785,791	24,100	29.81	30/30	1	0.404
PSO-VNS	736,832	787,709	19,268	30.25	30/30	2	–
PSO	729,189	787,989	21,150	30.98	27/30	3	8.49×10^{-4}
SSA-VNS	728,870	766,803	22,859	31.01	30/30	4	0.248
SSA	717,258	750,707	19,300	32.11	30/30	5	0.001
LX-SSA-VNS	715,952	757,838	30,626	32.23	29/30	6	0.002
LX-SSA	709,661	746,301	19,845	32.83	30/30	7	4.75×10^{-4}
RS-VNS	709,276	750,170	21,667	32.86	28/30	8	3.60×10^{-5}
MS-SLSQP	733,342	748,571	13,798	30.58	4/30	9	1.49×10^{-7}
DE	–	–	–	–	0/30	10	1.68×10^{-8}
<i>IEA37 CS1, 36 turbines, 30,030 calls</i>							
PSO-VNS	807,446	827,385	11,697	23.57	30/30	1	–
PSO	802,252	830,256	13,260	24.06	30/30	2	4.87×10^{-5}
VNS	801,797	824,230	11,876	24.10	30/30	3	0.119
SSA-VNS	791,404	819,931	14,356	25.09	30/30	4	4.18×10^{-4}
RS-VNS	787,892	807,208	9,532	25.42	30/30	5	4.16×10^{-6}
LX-SSA-VNS	785,677	810,989	13,216	25.63	30/30	6	1.12×10^{-7}
SSA	771,298	802,538	14,756	26.99	30/30	7	2.98×10^{-8}
LX-SSA	752,030	789,311	18,838	28.82	30/30	8	2.98×10^{-8}
MS-SLSQP	798,913	833,365	41,511	24.38	14/30	9	2.37×10^{-4}
DE	–	–	–	–	0/30	10	1.68×10^{-8}

Table S19: IEA Wind Task 37 Case Study 1: AEP (MWh, official calculator) of the baseline (example) layout, of the best feasible and the best overall participant layout, our best feasible layout over all methods and budgets, the best PSO-VNS layout, mean PSO-VNS AEP (budgets 6,030 / 30,030 calls), and difference (%) of the best PSO-VNS layout from the best feasible participant layout and from the baseline. “infeas.”: violates the boundary or spacing constraint by more than 1 mm.

Scenario	Baseline	Best feasible publ.	Best publ.	Our best	PSO-VNS best	PSO-VNS mean	Δ_{publ}	Δ_{base}
IEA37 CS1, 16 turbines	366,941.6	418,924.4 (par4)	421,561.9 (par12, infeas.)	413,331.9 (PSO-VNS)	413,331.9	395,031.8 / 402,583.1	-1.33	+12.64
IEA37 CS1, 36 turbines	737,883.1	863,676.3 (par4)	882,383.3 (par12, infeas.)	833,365.0 (MS-SLSQP)	827,385.0	736,832.4 / 807,446.0	-4.20	+12.13

Table S20: Robustness of the results to the benchmark model. All feasible final layouts of the 8 methods are re-evaluated (not re-optimized) with alternative power-curve and wake models. Δ : mean relative change of the objective (expected power) with respect to the benchmark model (both evaluated on the stored coordinates, rounded to 1 mm); average rank of each method over the 68 cases (ranking rule of Table I: methods with fewer than half of their runs feasible are ranked last, by their number of feasible runs); $\bar{\tau}$: mean Kendall rank correlation between the benchmark and the alternative ordering of the methods with at least 15 feasible runs within a case; “Same best”: percentage of cases in which the best method is unchanged.

Model	Δ (%)		Average rank										$\bar{\tau}$	Same best (%)
	DS I	DS II	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP				
Benchmark (linear curve, Jensen)	+0.0	+0.0	1.82	1.97	3.45	5.38	6.06	7.04	4.40	5.88	–	–		
Cubic power curve	-17.7	-36.9	1.82	1.90	3.53	5.68	5.97	6.90	4.42	5.79	0.95	94		
Cubic curve with 25 m/s cut-out	-21.5	-37.0	1.80	1.91	3.53	5.68	5.97	6.90	4.42	5.79	0.95	93		
Gaussian wake ($k^* = 0.04$)	-0.3	-0.4	1.97	1.91	3.93	5.43	6.04	6.94	5.12	4.66	0.66	54		

S-IV Wind Data

Table S21: Wind Data Sets I and II: probability p_l of direction bin l ($[15(l-1), 15l)^\circ$, direction the wind blows towards) and Weibull scale ψ ; $\zeta = 2$ in all bins.

Bin l	1	2	3	4	5	6	7	8	9	10	11	12
DS I: p_l ($\psi = 13$)	0.000	0.010	0.010	0.010	0.010	0.200	0.600	0.010	0.010	0.010	0.010	0.010
DS II: ψ	7.0	5.0	5.0	5.0	5.0	4.0	5.0	6.0	7.0	7.0	8.0	9.5
DS II: p_l	0.0002	0.0080	0.0227	0.0242	0.0225	0.0339	0.0423	0.0290	0.0617	0.0813	0.0994	0.1394
Bin l	13	14	15	16	17	18	19	20	21	22	23	24
DS I: p_l ($\psi = 13$)	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.000
DS II: ψ	10.0	8.5	8.5	6.5	4.6	2.6	8.0	5.0	6.4	5.2	4.5	3.9
DS II: p_l	0.1839	0.1115	0.0765	0.0080	0.0051	0.0019	0.0012	0.0010	0.0017	0.0031	0.0097	0.0317

S-V Comparison with Published Kusiak–Song Results

The Kusiak–Song benchmark has been solved with an evolution strategy [1], ant colony optimization [2], particle filtering [3] and biogeography-based optimization [4]. [TBD: author: literature table pending verification of the published values against the original papers (see analysis/literature_values.md)]

S-VI Additional Figures

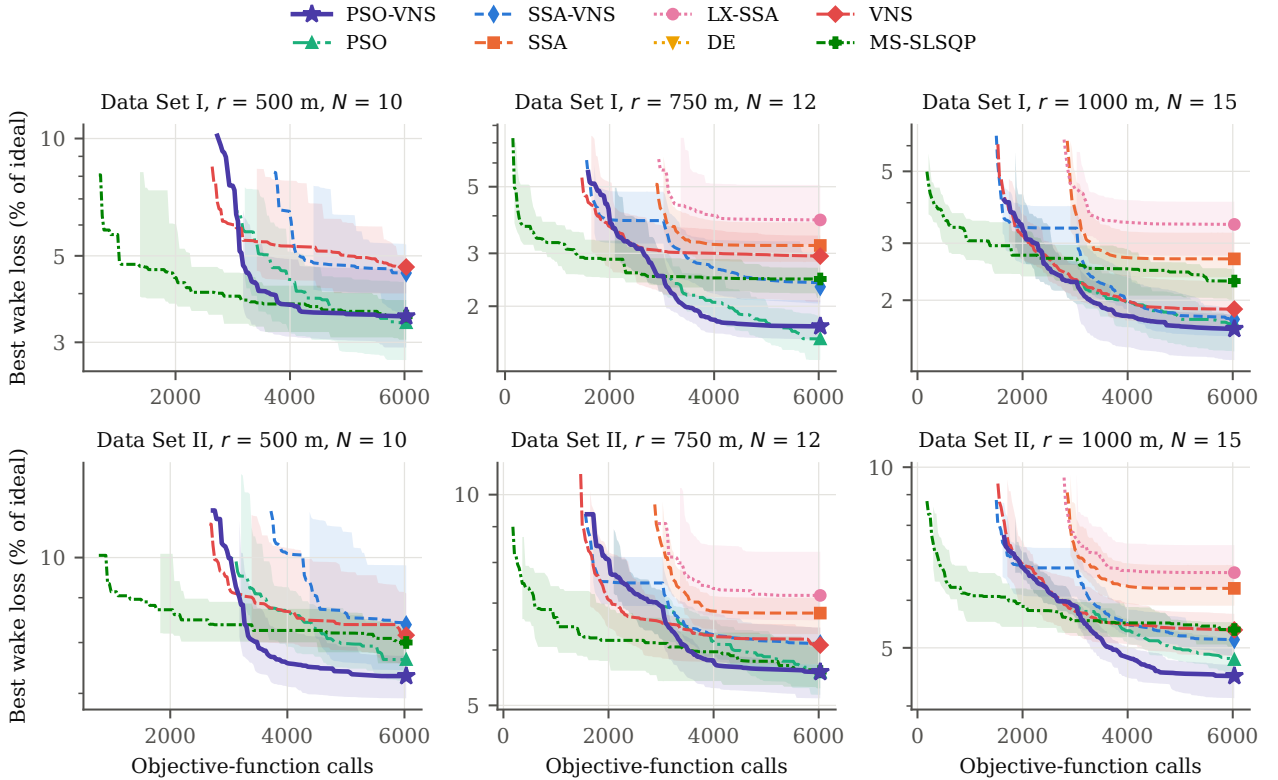


Figure S4: Convergence of the eight methods of the main comparison for the largest N of each farm: median (line) and interquartile range (band) of the best feasible wake loss over 30 runs versus objective calls.

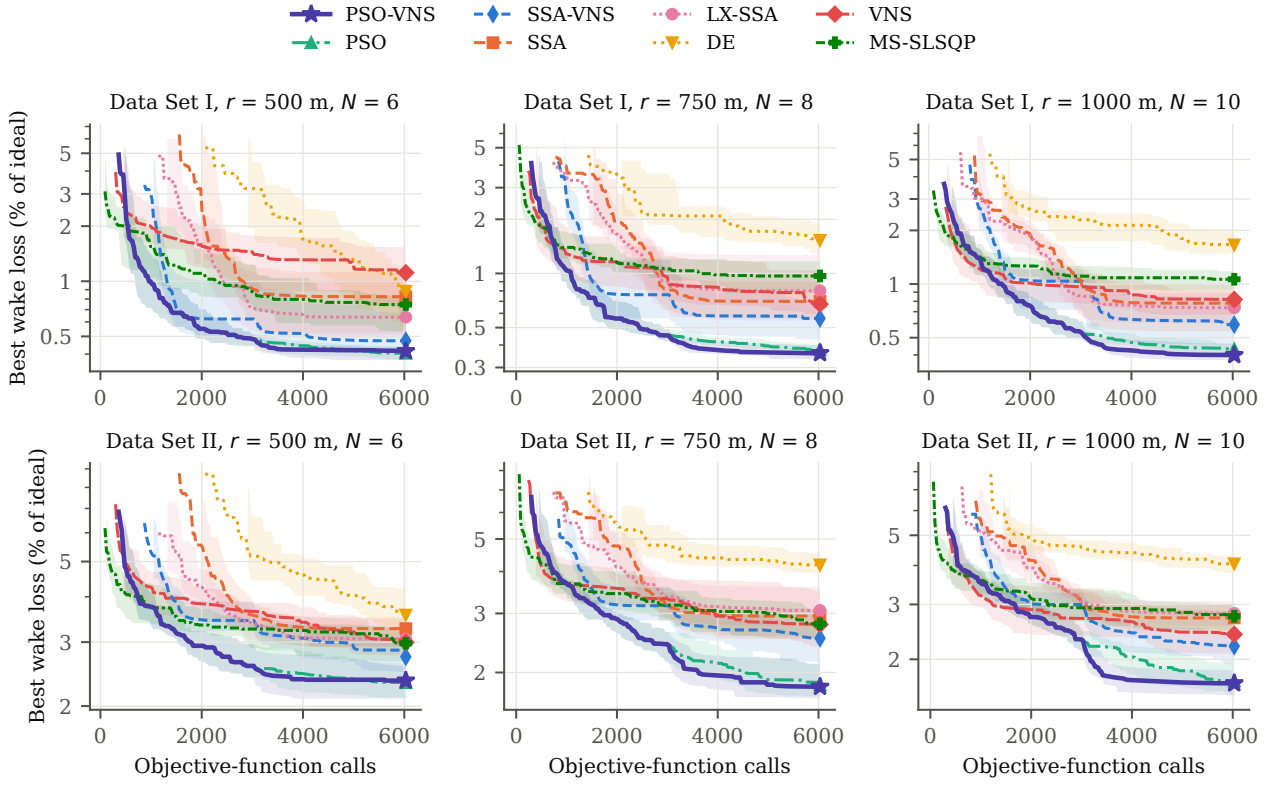


Figure S5: As Fig. S4, for the middle N of each farm ($N = 6, 8$ and 10 for $r = 500, 750$ and 1000 m).

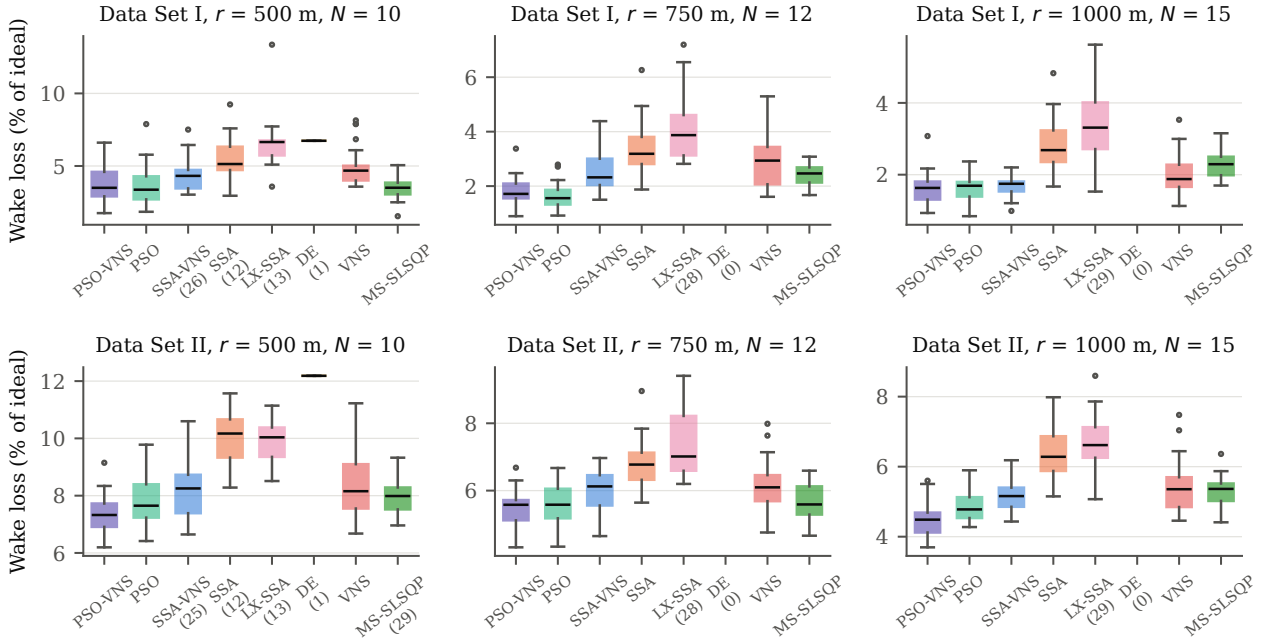


Figure S6: Distribution of the wake loss of the feasible runs for the largest N of each farm (number of feasible runs in parentheses when fewer than 30).

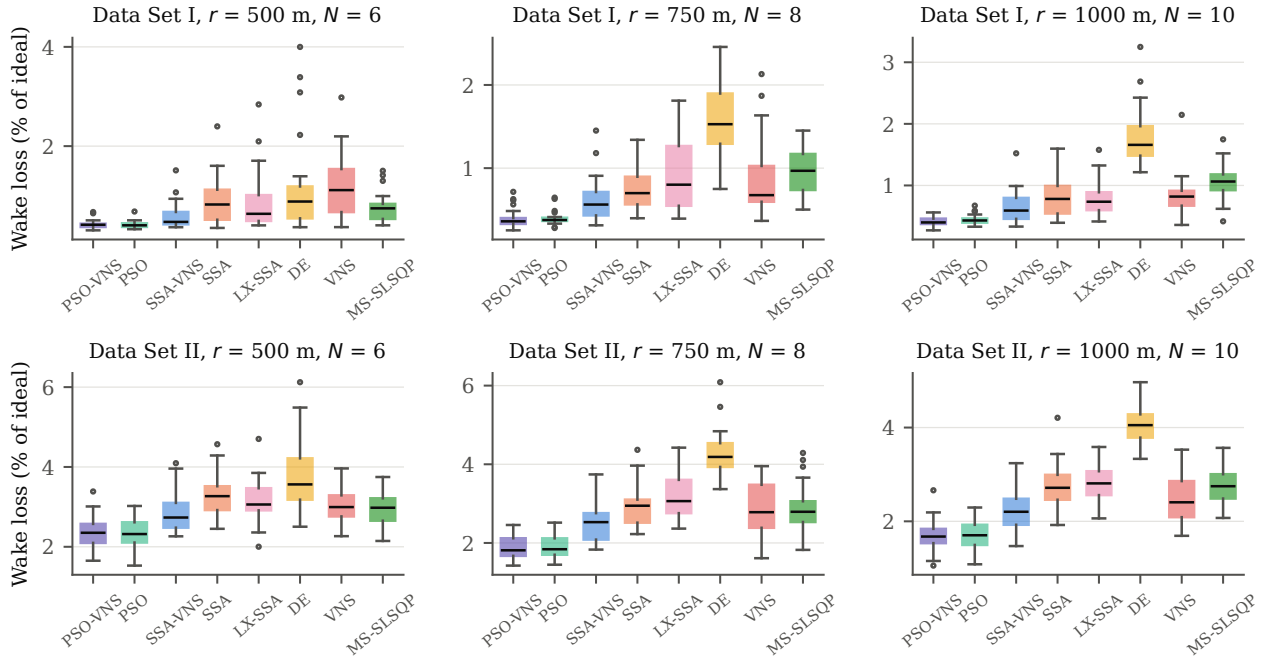


Figure S7: As Fig. S6, for the middle N of each farm.

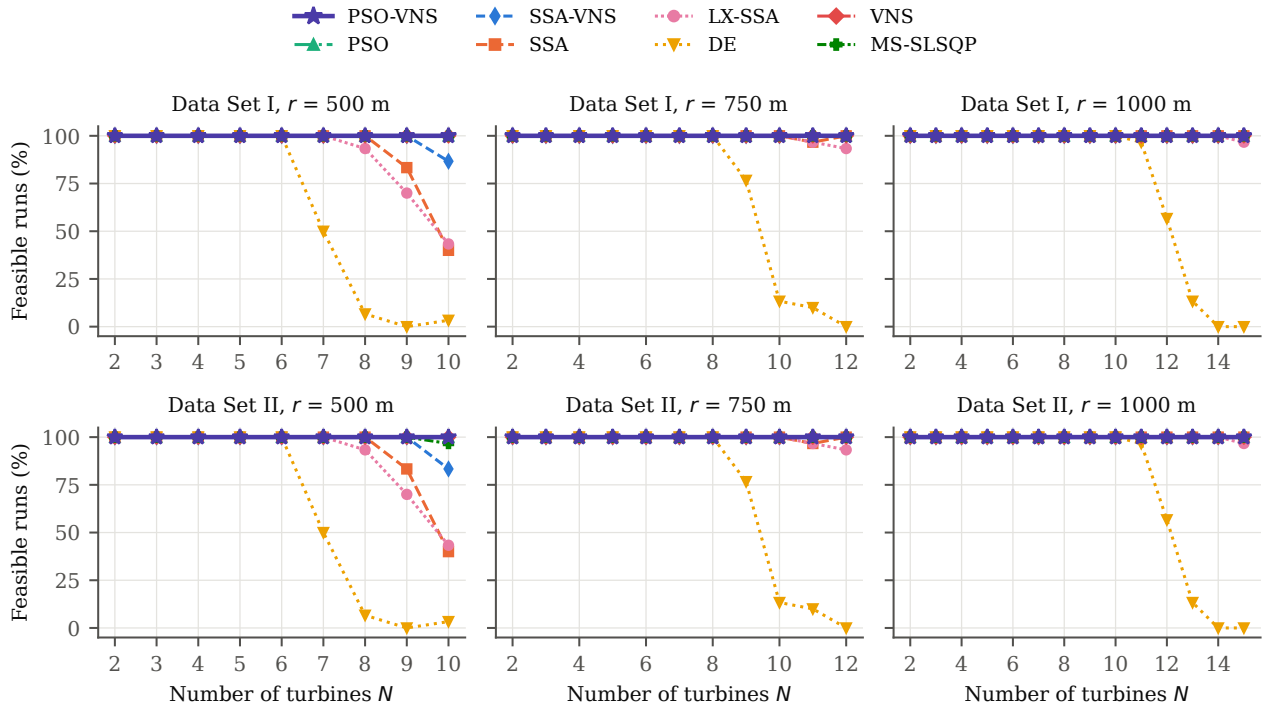


Figure S8: Percentage of feasible runs versus the number of turbines.

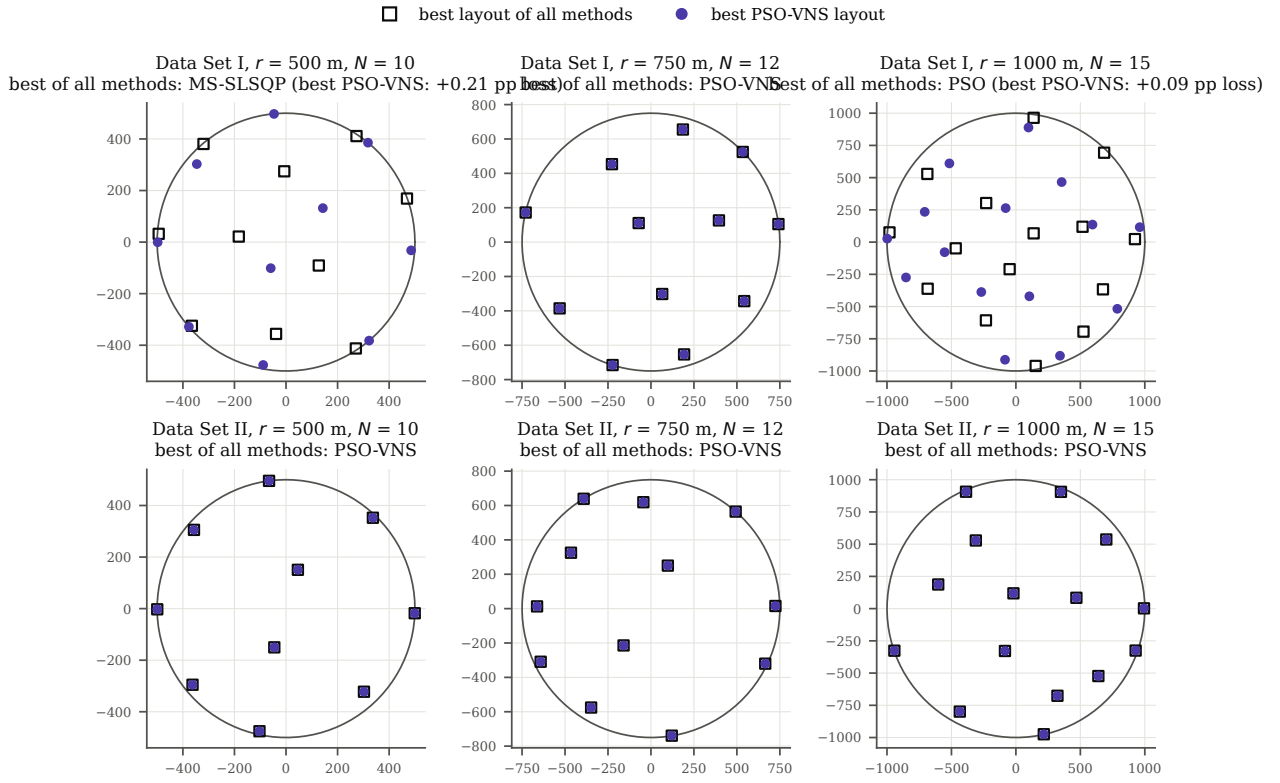


Figure S9: Best layout of all methods (squares) and best PSO-VNS layout (dots) for the largest N of each farm; the coordinates are listed in Table S10.

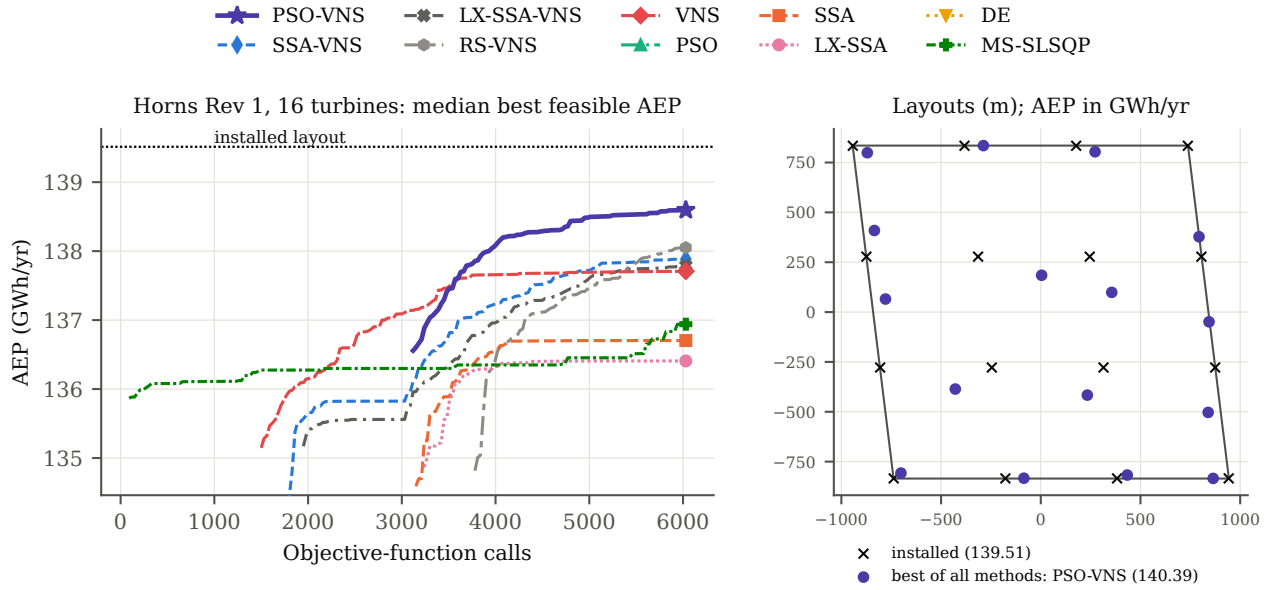


Figure S10: Horns Rev 1 16-turbine block: median best feasible AEP versus objective calls (left) and layouts (right).

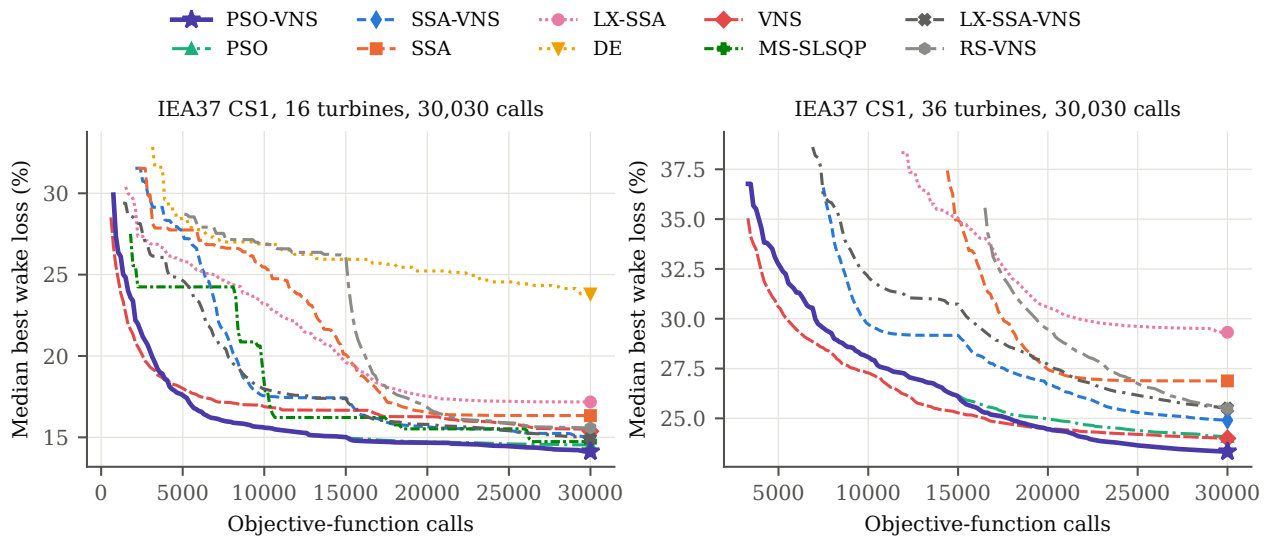


Figure S11: IEA37 Case Study 1: median best feasible wake loss versus objective calls at the largest budget.

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